**1.**

**(a)**

Show that the distance from the hyperplane $g(\mathbf{x}) = \mathbf{w^T x} + w_0 = 0$ to the point $\mathbf{x}_a$ is $|g(\mathbf{x}_a)|/\|\mathbf{w}\|$ by minimizing $\|\mathbf{x} - \mathbf{x}_a\|^2$ subject to the constraint $g(\mathbf{x}) = 0$.

We aim to show the statement by minimizing $\|\mathbf{x} - \mathbf{x}_a\|^2$ with $g(\mathbf{x}) = 0$. We build the corresponding Lagrangian which we can then attempt to minimize:

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda g(\mathbf{x})$$

$$= \|\mathbf{x} - \mathbf{x}_a\|^2 + \lambda \cdot (\mathbf{w^T x} + w_0)$$

We can cheat a bit and multiple the first term by $\frac{1}{2}$ to make taking the derivative and then solving easier; this is fine since we are solving for $\mathbf{x}$ rather than the value of the minimum itself, so we don't need to account for this later.

$$= \frac{1}{2} \cdot \|\mathbf{x} - \mathbf{x}_a\|^2 + \lambda \cdot (\mathbf{w^T x} + w_0)$$

Let's now take the derivative with respect to $\mathbf{x}$:

$$\frac{\partial}{\partial \mathbf{x}} L(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_a) + \lambda \mathbf{w}$$

and now setting to 0 to solve for the minimum:

$$0 = \mathbf{x} - \mathbf{x}_a + \lambda \mathbf{w}$$

$$\mathbf{x} = \mathbf{x}_a - \lambda \mathbf{w}$$

Plugging this back into the original hyperplane equation:

$$g(\mathbf{x}) = \mathbf{w^T x} + w_0 = 0$$

$$\mathbf{w^T}(\mathbf{x}_a - \lambda \mathbf{w}) + w_0 = 0$$

$$\mathbf{w^T x}_a - \lambda \mathbf{w^T w} + w_0 = 0$$

$$(\mathbf{w^T x}_a + w_0) - \lambda \|\mathbf{w}\|^2 = 0$$

$$g(\mathbf{x}_a) - \lambda \|\mathbf{w}\|^2 = 0$$

$$\lambda \|\mathbf{w}\|^2 = g(\mathbf{x}_a)$$
$$\lambda = \frac{g(\mathbf{x}_a)}{\|\mathbf{w}\|^2}$$

Looking back at our initial solution to the derivative of the Langrangian, from $\mathbf{x} = \mathbf{x}_a - \lambda\mathbf{w}$, we can see that $\mathbf{x} - \mathbf{x}_a = -\lambda\mathbf{w}$. So to find this distance, we just take the norm of this, i.e. $\|-\lambda\mathbf{w}\|$:

$$\begin{aligned} \|-\lambda\mathbf{w}\| &= \|\lambda\|\|\mathbf{w}\| \\ &= \frac{\|g(\mathbf{x}_a)\|}{\|\mathbf{w}\|^2} \cdot \|\mathbf{w}\| \\ &= \frac{\|g(\mathbf{x}_a)\|}{\|\mathbf{w}\|} = \frac{g(\mathbf{x}_a)}{\|\mathbf{w}\|} \end{aligned}$$

exactly as we sought to show.

**(b)**

Show that the projection of $\mathbf{x}_a$ onto the hyperplane is given by

$$\mathbf{x_p} = \mathbf{x}_a - \frac{g(\mathbf{x}_a)}{\|\mathbf{w}\|^2}\mathbf{w}$$

We kind of already showed this previously, with the last part of part (a) already using the equation

$$\mathbf{x} = \mathbf{x}_a - \lambda\mathbf{w}$$

and now taking the exact point on the hyperplane, we get

$$\mathbf{x}_p = \mathbf{x}_a - \lambda\mathbf{w}$$

and plugging in the value for $\lambda$ which we found previously:

$$\mathbf{x}_p = \mathbf{x}_a - \frac{g(\mathbf{x}_a)}{\|\mathbf{w}\|^2}\mathbf{w}$$

which is exactly what we sought to show.

**2.**

Let $\mathbf{x_1}, \ldots, \mathbf{x_n}$ be $n$ $q$-dimensional samples and $Q$ be any nonsingular positive definite $q \times q$ matrix. Show that the vector $\mathbf{x}$ that minimizes

$$\sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x})^T Q^{-1} (\mathbf{x_k} - \mathbf{x})$$

is the sample mean, $\bar{\mathbf{x}} = \frac{1}{n} \sum_{k=1}^{n} \mathbf{x_k}$.

We have the criterion function $J(a)$ as the exact equation we are minimizing, that is, $J(\mathbf{x}) = \sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x})^T Q^{-1} (\mathbf{x_k} - \mathbf{x})$. We therefore simply take the gradient and try to optimize accordingly:

$$\nabla_{\mathbf{x}} J(\mathbf{x}) = \sum_{k=1}^{n} 2Q^{-1}(\mathbf{x_k} - \mathbf{x}) \cdot (0 - 1)$$

taking the standard gradient and applying the chain rule. Continuing:

$$= \sum_{k=1}^{n} -2Q^{-1}(\mathbf{x_k} - \mathbf{x})$$

$$= -2Q^{-1} \sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x})$$

And optimizing by trying to solve for the minimum by setting to 0:

$$-2Q^{-1} \sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x}) = 0$$

$$\sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x}) = 0$$

$$\sum_{k=1}^{n} \mathbf{x_k} - \sum_{k=1}^{n} \mathbf{x} = 0$$

$$\sum_{k=1}^{n} \mathbf{x_k} - n\mathbf{x} = 0$$

$$n\mathbf{x} = \sum_{k=1}^{n} \mathbf{x_k}$$

$$\mathbf{x} = \frac{1}{n}\sum_{k=1}^{n}\mathbf{x_k}$$

which is indeed equivalent to the sample mean, as we sought to show.

**3.**

Consider a linear classifier with discriminant functions $g_i(\mathbf{x}) = \mathbf{w_i^T x} + w_{i0}, i = 1, \ldots, c$. Show that the decision regions are convex by showing that if $\mathbf{x}_1 \in \mathcal{R}_i$ and $\mathbf{x}_2 \in \mathcal{R}_i$ then $\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2 \in \mathcal{R}_i$ if $0 \leq \lambda \leq 1$.

By definition, we know that the linear machine will assign $\mathbf{x}$ to class $i$ if $g_i(\mathbf{x}) > g_j(\mathbf{x})$ for all $j \neq i$; as a byproduct of this, we also know that $g_i(\mathbf{x_1}) > g_j(\mathbf{x_1})$ and $g_i(\mathbf{x_2}) > g_j(\mathbf{x_2})$. As suggested by the problem statement, we aim to show that the decision regions are convex by showing that arbitrary point $\lambda\mathbf{x_1} + (1-\lambda)\mathbf{x_2}$ lies within the classification region with $0 \leq \lambda \leq 1$. We substitute this value for $\mathbf{x}$ into the linear classifier discriminant function:

$$g_i(\mathbf{x}) = \mathbf{w_i^T}(\lambda\mathbf{x_1} + (1-\lambda)\mathbf{x_2}) + w_{i0}$$

Looking at this – intuitively, we can see that this essentially is a complementary weighting of the two vectors since they sum to 1. We can therefore redistribute accordingly, with the bias distributed between the $\lambda$ and $1-\lambda$ weights, since this still adds up to $1 \cdot w_{i0}$:

$$= \lambda(\mathbf{w_i^T x_1} + w_{i0}) + (1-\lambda)(\mathbf{w_i^T x_2} + w_{i0})$$

and then substituting in the linear discriminant function definition accordingly:

$$g_i(\mathbf{x}) = \lambda g_i(\mathbf{x_1}) + (1-\lambda)g_i(\mathbf{x_2})$$

Based on the inequalities we previously established, we also know that $\lambda g_i(\mathbf{x_1}) \geq \lambda g_j(\mathbf{x_1})$, and $(1-\lambda)g_i(\mathbf{x_2}) \geq (1-\lambda)g_i(\mathbf{x_2})$, since multiplying by the $0 \leq \lambda \leq 1$ doesn't change the sign of the inequality but introduces the potential of possible equality in the case that either $\lambda$ or $1-\lambda$ is 0. Combining this with the actual

discriminant function that we found for $\mathbf{x}$, we see that we have

$$\lambda g_i(\mathbf{x_1}) + (1-\lambda) g_i(\mathbf{x_2}) > \lambda g_j(\mathbf{x_1}) + (1-\lambda) g_j(\mathbf{x_2})$$

with this being a strict inequality since only one of the weights being able to be 0, which is the only case of equality, thus leaving the other term to be a strict inequality. We already solved for this previously, so we are essentially just left with

$$g_i(\mathbf{x}) > g_j(\mathbf{x})$$

which shows us that this intermediate point satisfies the linear machine assignment rule that we first started with, which therefore indicates that indeed $\mathbf{x} \in \mathcal{R}_i$ as we hoped to show.

**4.**

In the gradient descent algorithm $\mathbf{a}_{k+1}$ is obtained from $\mathbf{a}_k$ by

$$\mathbf{a}_{k+1} = \mathbf{a}_k - \rho_k \nabla J(\mathbf{a}_k),$$

where $\rho_k$ is a positive scale factor that sets the step size. Consider the criterion function

$$J_q(\mathbf{a}) = \sum_{\mathbf{y} \in \mathcal{Y}} (\mathbf{a}^T \mathbf{y} - b)^2$$

where $\mathcal{Y}(\mathbf{a})$ is the set of samples for which $\mathbf{a}^T \mathbf{y} \leq b$. Suppose that $\mathbf{y}_1$ is the only sample in $\mathcal{Y}(\mathbf{a}_k)$. Show that $\nabla J_q(\mathbf{a}_k) = 2(\mathbf{a}_k^T \mathbf{y}_1 - b)\mathbf{y}_1$ and that the matrix of second partial derivatives is given by $D = 2\mathbf{y}_1 \mathbf{y}_1^T$. Use this to show that when the optimal $\rho_k$ is used in the gradient descent algorithm,

$$\mathbf{a}_{k+1} = \mathbf{a}_k + \frac{b - \mathbf{a}^T \mathbf{y}_1}{\|\mathbf{y}_1\|^2} \mathbf{y}_1.$$

Let's first calculate the gradient as the first thing that we want to show. Because we assume that $\mathbf{y}_1$ is the only sample, we have

$$J_q(\mathbf{a}_k) = (\mathbf{a}^T \mathbf{y}_1 - b)^2$$

and then taking the corresponding gradient:

$$\nabla J_q(\mathbf{a}_k) = 2(\mathbf{a}_k^T\mathbf{y}_1 - b) \cdot (\mathbf{y}_1 - 0)$$

$$= 2(\mathbf{a}_k^T\mathbf{y}_1 - b)\mathbf{y}_1$$

which matches what we expect.

Now, we take the second derivative to check the "matrix of second partial derivatives", first isolating $\mathbf{a}_k$ accordingly to try and get the correct form of the matrix:

$$\nabla J_q(\mathbf{a}_k) = 2(\mathbf{a}_k^T\mathbf{y}_1 - b)\mathbf{y}_1$$

$$= (2\mathbf{a}_k^T\mathbf{y}_1 - 2b)\mathbf{y}_1$$

$$= 2\mathbf{y}_1\mathbf{y}_1^T\mathbf{a}_k - 2b\mathbf{y}_1$$

and now taking the derivative again:

$$D = \frac{\partial}{\partial \mathbf{a}_k}\nabla J_q(\mathbf{a}_k) = 2\mathbf{y}_1\mathbf{y}_1^T - 0$$

$$= 2\mathbf{y}_1\mathbf{y}_1^T$$

which also matches what we aimed to show as the value for $D$.

We now move to deriving the update rule $\mathbf{a}_{k+1}$, first starting with finding the optimal step size $\rho_k$. By definition in the text, we have

$$\rho_k = \frac{\|\nabla J_q\|^2}{\nabla J_q^T H \nabla J_q}$$

for which $H$ is the Hessian, which in this case is the second partial derivative matrix $D$. Continuing onwards:

$$= \frac{(2(\mathbf{a}_k^T\mathbf{y}_1 - b)\mathbf{y}_1)^T(2(\mathbf{a}_k^T\mathbf{y}_1 - b)\mathbf{y}_1)}{(2(\mathbf{a}_k^T\mathbf{y}_1 - b)\mathbf{y}_1)^T(2\mathbf{y}_1\mathbf{y}_1^T)(2(\mathbf{a}_k^T\mathbf{y}_1 - b)\mathbf{y}_1)}$$

$$= \frac{4(\mathbf{a}_k^T\mathbf{y}_1 - b)^2\mathbf{y}_1^T\mathbf{y}_1}{8(\mathbf{a}_k^T\mathbf{y}_1 - b)^2\mathbf{y}_1^T\mathbf{y}_1\mathbf{y}_1^T\mathbf{y}_1}$$

$$= \frac{4(\mathbf{a}_k^T\mathbf{y}_1 - b)^2\|\mathbf{y}_1\|^2}{8(\mathbf{a}_k^T\mathbf{y}_1 - b)^2\|\mathbf{y}_1\|^4}$$

$$\rho_k = \frac{1}{2\|\mathbf{y}_1\|^2}$$

And finally plugging this back in:

$$\begin{aligned}
\mathbf{a}_{k+1} &= \mathbf{a}_k - \rho_k \nabla J(\mathbf{a}_k) \\
&= \mathbf{a}_k - (\frac{1}{2\|\mathbf{y}_1\|^2}) \cdot 2(\mathbf{a}_k^T \mathbf{y}_1 - b)\mathbf{y}_1 \\
&= \mathbf{a}_k - \frac{(\mathbf{a}_k^T \mathbf{y}_1 - b)\mathbf{y}_1}{\|\mathbf{y}_1\|^2} \\
\mathbf{a}_{k+1} &= \mathbf{a}_k + \frac{b - \mathbf{a}_k^T \mathbf{y}_1}{|\mathbf{y}\|^2}\mathbf{y}_1
\end{aligned}$$

which is exactly what we sought to show.

**5.**

Show that the partial derivatives of the functions $y_i = \exp(a_i)/\sum_k \exp(a_k)$ used in multiple class logistic discrimination are given by

$$\frac{\partial y_i}{\partial a_j} = y_i(\delta_{ij} - y_j)$$

where $\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & \text{otherwise.} \end{cases}$

We have that $y_i = \frac{\exp(a_i)}{\sum_k \exp(a_k)}$. Generally, we use the quotient rule here:

$$\frac{\partial y_i}{\partial a_j} = \frac{(\frac{\partial}{\partial a_j}\exp(a_i)) \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \frac{\partial}{\partial a_j}\sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2}$$

Now let's start case by case for the value of $\delta_{ij}$. For $i \neq j$:

$$\begin{aligned}
\frac{\partial y_i}{\partial a_j} &= \frac{0 \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \frac{\partial}{\partial a_j}\sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2} \\
&= \frac{-\exp(a_i) \cdot \frac{\partial}{\partial a_j}\sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2} \\
&= \frac{-\exp(a_i) \cdot \exp(a_j)}{(\sum_k \exp(a_k))^2}
\end{aligned}$$

$$= \frac{-\exp(a_i)}{\sum_k \exp(a_k)} \cdot \frac{\exp(a_j)}{\sum_k \exp(a_k)}$$

and substituting back based on the form of $y_i$ that we had before:

$$= -y_i y_j$$

$$\frac{\partial y_i}{\partial a_j} = y_i(0 - y_j)$$

which indeed fits the equation we aimed to find with the corresponding value of $\delta_{ij} = 0$. Now solving for when $i = j$:

$$\frac{\partial y_i}{\partial a_j} = \frac{\exp(a_i) \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \frac{\partial}{\partial a_j} \sum_k \exp(a_k)}{(\sum_k \exp(a_k))^2}$$

$$= \frac{\exp(a_i) \cdot (\sum_k \exp(a_k)) - \exp(a_i) \cdot \exp(a_j)}{(\sum_k \exp(a_k))^2}$$

$$= \frac{\exp(a_i) \cdot (\sum_k \exp(a_k))}{(\sum_k \exp(a_k))^2} - \frac{\exp(a_i) \cdot \exp(a_j)}{\sum_k \exp(a_k) \cdot \sum_k \exp(a_k)}$$

$$= \frac{\exp(a_i)}{\sum_k \exp(a_k)} - \frac{\exp(a_i)}{\sum_k \exp(a_k)} \cdot \frac{\exp(a_j)}{\sum_k \exp(a_k)}$$

and doing the same substitutions:

$$= y_i - y_i \cdot y_j$$

$$\frac{\partial y_i}{\partial a_j} = y_i(1 - y_j)$$

which also fits the equation we aimed to find with corresponding value $\delta_{ij} = 1$.

**6.**

In Module 1 we introduced nine (9) datasets. The data files are posted here in Module 9. Choose three (3) of these datasets and for each of the three:

(a)

i.

ii.

(a)

i.

ii.

iii.

iv.

A.